Accent Identification and Pronunciation Feedback from Frozen Self Supervised Speech Representations on Commodity Hardware

Sakrit Shramish
Independent Research
Email: shramish4@gmail.com

***Abstract*: Computer assisted pronunciation training systems typically depend on large labelled corpora and on graphics accelerators that place them outside the reach of ordinary deployment. This paper presents a conversational accent identification and pronunciation feedback system trained and served entirely on a commodity processor with no graphics accelerator, inside a training budget of approximately two hours. The design freezes a self supervised speech encoder, caches multi layer statistics pooled embeddings once, and spends the remaining budget on architecture search and seed ensembling rather than on a single partial fine tuning pass. Pronunciation quality is assessed independently of the accent model using goodness of pronunciation computed from connectionist temporal classification forced alignment, and the mapping from that quantity to a reported score is calibrated empirically against native speech rather than selected by inspection. An audit of the corpus revealed that every speaker in its published evaluation partition also appears in its training partition, which permits a model to score well by recognising voices rather than accents; the corpus is therefore repartitioned so that no speaker is shared across partitions. On this speaker disjoint partition the system attains a macro averaged F1 score of 0.4268 and an accuracy of 0.5034 across 23 accent classes, against a chance level of 0.0435. Analysis shows that first language accents are recovered reliably while native varieties of English are confused among themselves, with 89.1 percent of native variety recordings assigned to some native variety. End to end latency is approximately two seconds per utterance.**

***Index Terms*: accent identification, computer assisted pronunciation training, self supervised speech representation, goodness of pronunciation, forced alignment, speaker independent evaluation**

I. Introduction

Spoken language learners benefit from feedback that is specific, immediate and grounded in measurable properties of their speech. Two capabilities support such feedback. The first is accent identification, which situates a speaker relative to a population and indicates the first language interference patterns most likely to affect them. The second is pronunciation assessment, which localises difficulty to individual phonemes and words.

Most published systems for these tasks assume the availability of graphics accelerators and of many hours of labelled speech. That assumption excludes deployment on ordinary personal machines and raises the cost of experimentation. This paper asks whether a useful system of this kind can be constructed under three simultaneous constraints: a processor only environment, a training budget of a small number of hours, and a requirement that all reported scores be computed by measurement rather than generated by a language model.

The contributions of this work are as follows.

1) A processor only pipeline that freezes a self supervised encoder and caches multi layer statistics pooled embeddings, converting the training budget into architecture search and ensembling. A measured optimisation raises encoder throughput from 10.0 to 22.5 times real time.
2) An audit of the accent corpus used here showing complete speaker overlap between its published training and evaluation partitions, together with a speaker disjoint repartitioning that restores the validity of the evaluation.
3) An empirical calibration of the goodness of pronunciation scale against measured native speech, replacing a constant chosen by inspection that misreported native speakers by more than forty points.
4) An analysis separating two distinct error regimes, namely reliable recovery of first language accents and systematic confusion among native varieties of English.

II. Related Work

Self supervised speech representation learning produces encoders whose intermediate activations transfer to many downstream tasks. Wav2vec 2.0 [1] established contrastive pretraining over masked latent speech representations, and WavLM [2] extended the objective with denoising and speaker discriminative components, improving performance on tasks that depend on speaker and channel properties. Benchmarks over frozen encoders [3] established that a learned weighted

TABLE I
CLASS DISTRIBUTION OF THE CORPUS

Frequency	Accent classes and recording counts
Above 2500	English 14252, American 8779, Scottish 7928, Irish 4394, Canadian 3151
500 to 2600	NorthernIrish 2581, Indian 1434, Spanish 1259, Dutch 1202, German 1113, Czech 974, Polish 907, French 793, Italian 760, Hungarian 676, Vietnamese 590, Finnish 590, Romanian 586
Below 500	Slovak 409, Estonian 289, Lithuanian 178, Croatian 127, Slovene 66

combination of transformer layers, followed by a light prediction head, is a strong and inexpensive way to exploit such encoders, and that the informative layers differ by task.

Pronunciation assessment has long been organised around the goodness of pronunciation measure introduced by Witt and Young [4], which compares the posterior probability of an intended phone against the best competing phone under an acoustic model. Contemporary implementations obtain the required frame level posteriors from connectionist temporal classification models [5] rather than from hidden Markov model systems, and obtain the intended phone sequence by grapheme to phoneme conversion of a recognised transcript.

Accent identification is frequently evaluated on corpora in which each speaker carries a single accent label. Under that condition an evaluation partition sharing speakers with the training partition measures speaker recognition in addition to accent recognition. Speaker disjoint partitioning is therefore standard practice in speaker attribute research, and the present work applies it to a corpus whose distributed partitions do not satisfy it.

III. DATASET

A. Source and Composition

Experiments use the English Accent DataSet distributed on the HuggingFace Hub under the identifier `westbrook/English_Accent_DataSet`. Sample identifiers indicate that the audio derives from the Edinburgh International Accents of English Corpus [6], a collection of spontaneous conversational English. Conversational rather than read speech is appropriate here because the intended application is a spoken dialogue system.

The corpus contains 53,038 recordings from 519 speakers, totalling 75.1 hours at a sampling rate of 16 kHz, and supplies an accent label drawn from 23 classes, a speaker identifier, a reference transcript, speaker gender and duration. The mean recording length is 5.4 seconds and the median is 3.6 seconds. Table I summarises the class distribution, which spans more than two orders of magnitude.

Because class frequency is so uneven, overall accuracy is an unsuitable primary metric: a model that ignored every infrequent accent could still report a high value. Macro averaged F1, which weights all 23 classes equally, is therefore adopted as the primary metric, with balanced accuracy reported alongside it.

TABLE II
SPEAKER OVERLAP BETWEEN PUBLISHED PARTITIONS

Partition pair	Shared speakers	Proportion
Training and test	250 of 250	1.000
Training and validation	230 of 232	0.991
Validation and test	199	

TABLE III
SPEAKER DISJOINT PARTITIONS USED IN THIS WORK

Partition	Clips	Speakers	Hours	Class cap	Speaker cap
Training	5445	299	7.45	480	70
Validation	837	73	1.09	110	30
Test	890	91	1.16	110	30
Shared speakers between any two partitions: 0					

B. Speaker Leakage in the Published Partitions

Before any model was trained, the published partitions were audited for speaker overlap. The result is reported in Table II.

Every speaker in the published test partition also appears in the published training partition. Since each speaker carries exactly one accent label, a model can attain a high score on that partition by associating a voice with its label, without acquiring any transferable representation of accent. Results obtained on the published partitions are therefore optimistic by an unknown margin and do not predict behaviour on an unseen user, which is precisely the condition under which the intended application operates.

C. Speaker Disjoint Repartitioning

All recordings were pooled and repartitioned by speaker. Speakers were allocated to partitions independently within each accent class, so that every class remains represented in all three partitions; all 23 classes contain at least six distinct speakers, making this feasible throughout.

Two sampling limits are applied. A per class limit compresses the frequency imbalance. A per speaker limit is equally important and less obvious: a single Lithuanian speaker contributes 140 of that class's 178 recordings, so without such a limit the class degenerates into a single voice prototype and the model learns that speaker rather than the accent. Recordings are drawn in rotation across the available speakers so that voice coverage within a class remains even. Table III reports the resulting partitions. Speaker disjointness was asserted programmatically and holds for all three partition pairs.

IV. METHODOLOGY

A. System Architecture

The system processes one spoken utterance per turn. Fig. 1 shows the arrangement. Recognition runs first because the

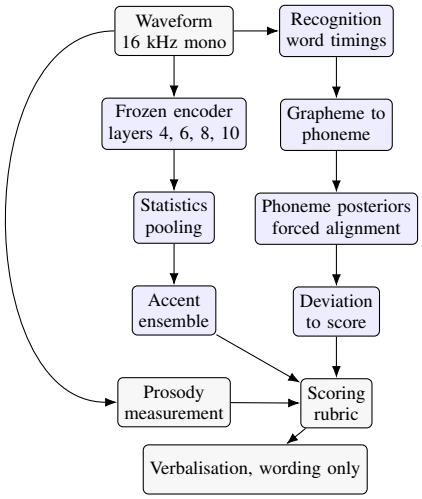


Fig. 1. System arrangement. Shaded blocks carry learned components. Every reported score originates in the rubric; the language model receives those values already computed and contributes wording alone.

pronunciation branch requires a reference word sequence; the accent branch and the prosody branch operate directly on the waveform and are independent of recognition output. Scores from the three branches are combined by a fixed rubric, and only the final verbalisation step is delegated to a language model.

B. Frozen Encoder with Multi Layer Statistics Pooling

The absence of a graphics accelerator makes end to end fine tuning of a 94 million parameter encoder impractical within the stated budget. The encoder is therefore frozen and used purely as a feature extractor, and its output is cached once. A single training run over cached features then costs seconds rather than hours, which converts the remaining budget into many epochs, an architecture search and a seed ensemble. That exchange is the central design decision of this work.

The encoder is WavLM Base Plus [2]. Rather than reading only its final layer, activations are taken from transformer layers 4, 6, 8 and 10, following the observation that intermediate layers of self supervised speech encoders carry the most phonetic information while upper layers specialise toward the pretraining objective [3]. Layers above the highest index used are removed from the computation graph entirely, which is a correctness preserving saving of roughly one sixth of encoder cost.

For layer ℓ with frame activations $h_t^{(\ell)} \in \mathbb{R}^{768}$ and valid frame set F determined by the attention mask, the pooled representation concatenates the elementwise mean

$$\mu^{(\ell)} = \frac{1}{|F|} \sum_{t \in F} h_t^{(\ell)} \tag{1}$$

with the elementwise standard deviation of $h_t^{(\ell)}$ over the same frame set. The standard deviation term is what preserves articulatory variation across the utterance, which mean pooling alone discards. Concatenating the four layers yields an utterance embedding of $4 \times 2 \times 768 = 6144$ dimensions.

Masking is applied throughout pooling so that padded frames contribute nothing. This was verified by encoding a recording alone and again inside a padded batch; the two embeddings agree to within 2.4 parts in ten million, confirming that batch composition does not influence the representation.

C. Accent Classification Head

Two heads were considered. The first applies a learned convex combination over the four cached layers, with weights

$$\alpha = \mathrm{softmax}(w), \qquad z = \sum_{\ell} \alpha_\ell \, e^{(\ell)}, \tag{2}$$

followed by layer normalisation and a two hidden layer perceptron of widths 512 and 256 with Gaussian error linear units and dropout. The second is a linear probe applied to the full 6144 dimensional concatenation, preceded by layer normalisation and dropout.

Optimisation uses decoupled weight decay [11] under a one cycle schedule [12], with label smoothing, optional embedding mixup [13] and class weights proportional to the square root of inverse class frequency so that infrequent accents are not discarded in favour of frequent ones. Checkpoints are selected on validation macro F1.

The mixing weights w require separate treatment. Their gradients are roughly two orders of magnitude smaller than those reaching the perceptron, because the classifier can compensate internally for a poor mixture. Under a shared step size they never depart from their uniform initialisation, and the learned mixture would be decorative rather than informative. They are therefore assigned a step size forty times larger and exempted from weight decay, since decay actively pulls the mixture back toward uniformity.

D. Pronunciation Assessment

Pronunciation quality is not learned from the accent corpus and is not inferred by a language model. It is computed from acoustic posteriors following the goodness of pronunciation formulation [4].

A wav2vec 2.0 Base model fine tuned on the TIMIT corpus [7] for ARPAbet phoneme recognition supplies frame level posteriors over 39 phone classes. This inventory was chosen deliberately: it corresponds one to one with the CMU Pronouncing Dictionary [8], so the reference and hypothesis share a symbol set and no lossy mapping is required. The reference phone sequence is obtained by grapheme to phoneme conversion of the recognised transcript, and connectionist temporal classification forced alignment [5] assigns each reference phone a frame span T_p.

For phone p with competing inventory Q, the deviation is the mean log ratio between the most probable phone and the intended phone across that span,

$$D(p) = \frac{1}{|T_p|} \sum_{t \in T_p} \log \frac{\max_{q \in Q} P(q \mid x_t)}{P(p \mid x_t)}. \tag{3}$$

A deviation of zero indicates that the intended phone was also the most probable phone over its span; larger values indicate that the acoustic evidence favours some other phone. Equation (3) is the standard goodness of pronunciation quantity expressed with the sign convention reversed, so that the reported quantity increases with error.

The deviation is mapped onto a bounded score by

$$S(p) = \frac{100}{\exp\bigl(D(p)/k\bigr)}, \tag{4}$$

and phone scores are aggregated into articulatory groups, such as dental fricatives, rhotic approximants and tense against lax vowel pairs, so that feedback identifies a practicable articulatory target rather than an isolated symbol.

Word level reporting requires care in two respects. First, each word must carry its own phones through the alignment rather than being matched positionally against the recogniser output, because words absent from the dictionary produce no phones and are dropped, which causes the two sequences to drift and attributes low scores to the wrong words. Second, closed class function words are excluded from reported practice targets. Such words reduce heavily in natural speech, so they score poorly against dictionary citation forms for every speaker including natives, and presenting them as practice targets conveys no instructional information.

E. Calibration of the Pronunciation Scale

The constant k in (4) sets the entire reported scale and cannot responsibly be chosen by inspection. The acoustic model is trained on read American English while the evaluation corpus is spontaneous conversational speech of many varieties, so absolute deviations are systematically larger than intuition suggests. An initial value selected by inspection assigned a native speaker a score of 42 out of 100, which is not a defensible report.

The constant is therefore measured. Deviation is computed over n recordings by speakers of native English varieties, using the corpus reference transcript rather than recogniser output so that the measurement reflects pronunciation alone and is not contaminated by recognition error. Writing D_{50} for the median deviation over that sample and $S^{\star}$ for the score that the median native speaker should receive, (4) gives

$$k = \frac{D_{50}}{\ln\bigl(100/S^{\star}\bigr)}. \tag{5}$$

A reported score then carries an interpretable meaning, namely position relative to measured native speech, rather than being an arbitrary function of a log ratio.

F. Fluency and Prosody Measurement

Fluency and prosody quantities are obtained by deterministic signal processing, without any model, so that each is reproducible and explainable to a learner. Voice activity is detected by an adaptive energy threshold on 25 millisecond frames at a 10 millisecond hop, set relative to the loud and quiet spread of the utterance itself so that it adapts to recording level. From the resulting activity pattern the system derives pause count, total and longest pause duration, and pause proportion. Speaking rate and articulation rate, the latter computed over speech intervals only, are derived from a syllable estimate over recognised words. Fundamental frequency is tracked between 60 and 400 Hz and converted to semitones about its own median, which yields a pitch range that is speaker normalised and therefore comparable across voices. Each quantity is mapped to a bounded subscore by a monotonic function of a single measured variable, so that any reported value can be traced to the evidence producing it.

TABLE IV
MEASURED ENCODER THROUGHPUT

Configuration	Throughput	Note
Twelve layers, batch 1	10.0	naive baseline
Twelve layers, batch 16	16.7	batching alone
Ten layers, batch 16	18.4	unused layers removed
First deployment	6.0	below benchmark
Cross group buffering	22.5	final configuration

G. Response Generation

A language model is used for one purpose, namely phrasing the reply that the learner reads. It receives quantities that have already been computed and is instructed not to originate, revise or contradict any of them. The deployed configuration uses a deterministic writer driven by the same quantities, so the system operates with no external service. This separation means that the measurement pipeline is fully responsible for every reported number, and that enabling a hosted model changes wording alone.

V. EXPERIMENTAL SETUP

A. Computational Environment

All work was performed on a single consumer laptop with an eight core processor exposing twelve hardware threads, 15 GB of memory and no graphics accelerator of any kind. Software comprised PyTorch 2.14 built for processors only, Transformers 5.17 and a connectionist temporal classification inference runtime for recognition. No hosted inference service was used at any stage of training or evaluation.

B. Encoder Throughput Optimisation

Since embedding extraction dominates total cost, encoder throughput was measured rather than assumed. Table IV reports the results in multiples of real time, that is seconds of audio processed per second of computation.

The gap between the 18.4 benchmark and the 6.0 initially observed in deployment was diagnosed rather than accepted. Each storage row group contained only about 25 wanted recordings, so batches were both small and heavily padded; because a batch costs as much as its longest member, combining a 1.5 second recording with an 8 second recording wastes most of the computation. Buffering 384

TABLE V
ACCENT IDENTIFICATION OVER 23 CLASSES

Metric	Validation	Test	Chance
Macro averaged F1	0.4575	**0.4268**	0.0435
Accuracy	0.5579	**0.5034**	0.0435
Balanced accuracy	0.4856	**0.4549**	0.0435

recordings across row groups and sorting globally by length before batching produces nearly uniform batches and raises throughput beyond the fixed length benchmark. Full extraction then processed 9.69 hours of audio in 2212 seconds, an average of 15.8 times real time including all decoding and input overhead.

A second measured constraint concerned data transfer. Acquisition became network bound, and observed throughput fell to roughly 250 kB per second while the processor stood idle. Comparison against an unrelated host showed the connection itself to be healthy, which localised the fault to a prefetch queue that stopped requesting data whenever the encoder was busy. Moving acquisition into an independent process restored 2.2 MB per second and allowed transfer and computation to proceed concurrently.

C. Training Protocol

Because features are cached, a complete training run costs seconds. The budget was therefore allocated to a search over 24 configurations spanning both head architectures, hidden widths, dropout rates, learning rates and the presence of mixup. Screening runs use a reduced epoch count, since ranking stabilises well before convergence, and the selected configuration is retrained under the full schedule. Seven independently seeded models are then trained under that configuration and their posterior distributions averaged.

Model selection used the speaker disjoint validation partition exclusively. The test partition was evaluated once, after selection was complete.

VI. RESULTS

A. Accent Identification

Table V reports the principal result. All values are obtained on the speaker disjoint partition, so no speaker contributing to evaluation was heard during training.

The selected configuration was the linear probe over the full concatenation of cached layers, which outperformed every perceptron variant in the search. This outcome justifies including a linear baseline rather than presuming that the larger head would prevail: the probe observes all four layers at full dimensionality, whereas the perceptron first compresses them into a single weighted combination and so discards information before classification. The seven ensemble members attained test macro F1 between 0.3890 and 0.4463, and their averaged posteriors attained 0.4268, within the spread of the members.

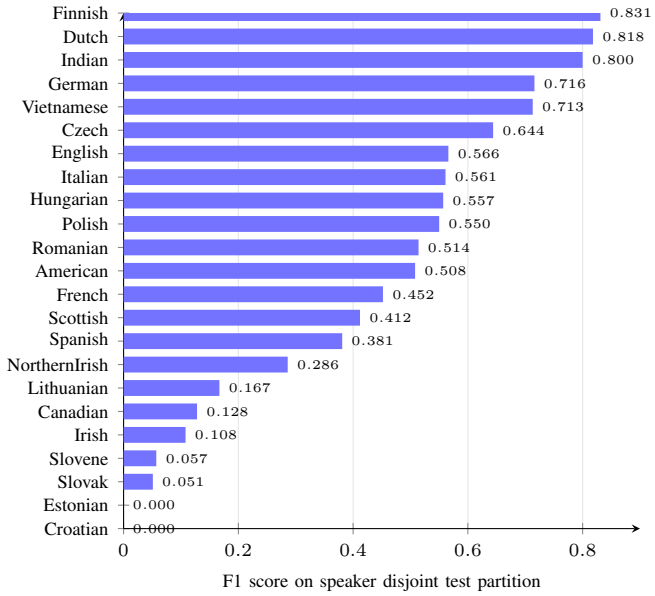


Fig. 2. Per class F1 on the speaker disjoint test partition. First language accents occupy the upper region; native varieties of English and classes with very few speakers occupy the lower region.

B. Per Class Behaviour

Fig. 2 reports per class F1 on the test partition. Performance separates sharply into regimes that differ in kind rather than degree.

First language accents are recovered reliably, with Finnish at 0.831, Dutch at 0.818, Indian at 0.800, German at 0.716 and Vietnamese at 0.713. The lower region contains two distinct populations. The first comprises native varieties of English, namely Irish at 0.108, Canadian at 0.128 and Northern Irish at 0.286, which are analysed in Section VII. The second comprises classes with very little material, namely Estonian with 19 test recordings, Croatian with 6 and Slovene with 3, for which a speaker disjoint partition leaves too little evidence to support a stable estimate.

C. Encoder Layer Contribution

The learned mixture over cached encoder layers, recovered from the perceptron arm of the search, assigns 37.3 percent to layer 4, 35.7 percent to layer 6, 15.5 percent to layer 8 and 11.5 percent to layer 10. Almost three quarters of the mixture therefore falls on the two lowest layers examined, supporting the view that accent is carried predominantly by lower and middle encoder representations and is progressively discarded in upper layers. This measurement was only obtainable after the mixing weights were given an independent step size; under a shared step size they remained at their uniform initialisation and reported nothing.

D. Calibration of the Pronunciation Scale

Calibration over 100 native English recordings yields a median deviation of 1.519 and therefore a constant of $k =$

TABLE VI
CALIBRATED PRONUNCIATION SCALE OVER NATIVE SPEECH

Percentile of native speech	Deviation	Score
5th	3.115	73.4
25th	2.208	80.2
50th	1.519	86.0
75th	1.067	90.0
95th	0.474	95.4

TABLE VII
CONFUSION AMONG NATIVE VARIETIES ON THE TEST PARTITION

True	Eng	Sco	Iri	NIr	Ame	Can
English	69	15	6	0	3	0
Scottish	41	34	9	4	5	3
Irish	14	6	5	2	12	11
NorthernIrish	6	1	1	6	8	7
American	3	6	1	0	31	6
Canadian	1	1	0	0	8	3

10.07, against a value of 3.2 chosen initially by inspection. Table VI reports the resulting scale.

Native speech now occupies a band between roughly 73 and 95, with the median at 86, and non native speech falls below that band. The correction is substantial: under the value chosen by inspection, the median native speaker would have been reported at approximately 42 out of 100. This supports the general position that the mapping from an unbounded acoustic quantity to a learner facing score should be measured against a reference population rather than selected by judgement.

E. Latency

End to end processing of a single utterance, comprising recognition, forced alignment, pronunciation scoring, accent inference, prosody measurement and response construction, completes in approximately 1.5 to 2.5 seconds on the target processor with models resident in memory. This is compatible with conversational interaction. Model loading is performed once at service start so that the first learner utterance does not absorb initialisation cost.

VII. DISCUSSION

A. Confusion Structure Among Native Varieties

The weak per class scores for native varieties do not indicate that the model fails on those recordings. Table VII presents the confusion submatrix restricted to the six native varieties.

Across these six classes, 89.1 percent of recordings receive a prediction that is itself a native variety. The model therefore separates native English from first language accented English reliably, and the residual error is concentrated within the native group. That structure is linguistically coherent. Irish, Scottish and Northern Irish varieties share substantial phonological ground, and the Canadian and American distinction rests on a small number of cues that are frequently absent from a short conversational utterance. A practical consequence follows for the application: reporting a coarse grouping, or reporting the posterior mass assigned to native varieties collectively, conveys more reliable information to a learner than forcing a single fine grained label.

Confidence behaviour supports the same reading. The mean maximum posterior over the test partition is 0.359, so the model is explicitly uncertain rather than confidently wrong, and the interface reports that uncertainty. An inference path check confirmed that predictions computed through the live service reproduce those computed over cached embeddings exactly, so the deployed system does not deviate from the evaluated system.

B. Limitations

Four limitations bound these results.

First, training used a subset of the corpus. Acquisition bandwidth rather than computation was the binding constraint during the experimental window, so 12 of 30 storage shards were used, yielding 5445 training recordings from 299 speakers. Classes whose material is concentrated in unused shards are correspondingly weak, and additional material is expected to benefit them most.

Second, three classes have too few test recordings for a stable estimate. Slovene contributes 3 test recordings, Croatian 6 and Lithuanian 8. Reported F1 for these classes should be read as an indication of scarcity rather than as a measurement of capability.

Third, the pronunciation acoustic model is trained on read American English. Deviations computed for speakers of other native varieties therefore reflect distance from that reference as well as pronunciation difficulty. Calibration against native speech corrects the location of the scale but does not remove the reference dependence, and a system intended to be neutral across varieties would require either a variety matched acoustic model or a per variety calibration.

Fourth, the accent label space inherits the composition of the corpus, which is dominated by European first languages and native varieties of English. Speakers whose first language lies outside that space cannot be represented correctly and will be assigned the nearest available class.

VIII. CONCLUSION

This paper described an accent identification and pronunciation feedback system constructed entirely on a commodity processor without any graphics accelerator, within a training budget of roughly two hours. Freezing a self supervised encoder and caching multi layer statistics pooled embeddings reduced each training run to seconds, which allowed the budget to be spent on architecture search and seed ensembling rather than on a single partial fine tuning pass. On a speaker disjoint partition of a 23 class corpus the system attains a macro averaged F1 of 0.4268 and an accuracy of 0.5034 against a chance level of 0.0435, and serves a complete assessment in approximately two seconds.

Three findings generalise beyond this system. An audit of the corpus revealed that all 250 speakers in its published

evaluation partition also appear in its training partition, which permits apparent accuracy to be obtained by voice recognition; any evaluation of speaker attributes should verify speaker disjointness before reporting. Empirical calibration of the pronunciation scale against measured native speech changed the constant from 3.2 to 10.07 and moved the median native speaker from 42 to 86, showing that a mapping from an unbounded acoustic quantity to a learner facing score must be measured rather than chosen. Finally, the learned mixture over encoder layers assigned nearly three quarters of its mass to the two lowest layers examined, but only became informative once the mixing parameters were given a step size appropriate to their gradient scale; a learned combination that is optimised carelessly will report its initialisation and appear to be a result.

Future work will extend training to the full corpus, add a variety matched or per variety calibrated acoustic model for pronunciation scoring, and report coarse native groupings alongside fine grained labels so that the confidence the system reports matches the reliability it possesses.

References

[1] A. Baevski, Y. Zhou, A. Mohamed and M. Auli, “wav2vec 2.0: A framework for self supervised learning of speech representations,” in *Advances in Neural Information Processing Systems*, 2020.

[2] S. Chen *et al.*, “WavLM: Large scale self supervised pre training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, 2022.

[3] S. Yang *et al.*, “SUPERB: Speech processing universal performance benchmark,” in *Proc. Interspeech*, 2021.

[4] S. M. Witt and S. J. Young, “Phone level pronunciation scoring and assessment for interactive language learning,” *Speech Communication*, vol. 30, no. 2, 2000.

[5] A. Graves, S. Fernández, F. Gomez and J. Schmidhuber, “Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks,” in *Proc. International Conference on Machine Learning*, 2006.

[6] R. Sanabria, N. Bogoychev, N. Markl, A. Carmantini, O. Klejch and P. Bell, “The Edinburgh International Accents of English Corpus: Towards the democratization of English ASR,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, 2023.

[7] J. S. Garofolo *et al.*, “TIMIT acoustic phonetic continuous speech corpus,” Linguistic Data Consortium, 1993.

[8] Carnegie Mellon University, “The CMU Pronouncing Dictionary, version 0.7b,” 2014.

[9] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey and I. Sutskever, “Robust speech recognition via large scale weak supervision,” in *Proc. International Conference on Machine Learning*, 2023.

[10] D. Snyder *et al.*, “X vectors: Robust DNN embeddings for speaker recognition,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, 2018.

[11] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. International Conference on Learning Representations*, 2019.

[12] L. N. Smith and N. Topin, “Super convergence: Very fast training of neural networks using large learning rates,” in *Proc. SPIE Artificial Intelligence and Machine Learning for Multi Domain Operations Applications*, 2019.

[13] H. Zhang *et al.*, “mixup: Beyond empirical risk minimization,” in *Proc. International Conference on Learning Representations*, 2018.